Project: Reinforcement Learning & ASP Integration

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1 The Problem

The integration of artificial intelligence agents into human-centered environments represents a major challenge in contemporary computer science. As intelligent systems become increasingly embedded in everyday contexts, it is essential to design agents that not only optimize formal objectives but also interact effectively with humans, accounting for their emotional and behavioral dynamics.

This project addresses this challenge by proposing a model that combines Reinforcement Learning (RL) with Answer Set Programming (ASP). The goal is to design an agent capable of interacting with humans and adapting its behavior based on observed emotional states, aiming to maximize overall happiness while minimizing negative or harmful emotional outcomes.

Several intrinsic challenges arise in this context. Human reactions, while often consistent to some extent, are highly individual: different people may respond differently to the same stimuli. Additionally, the surrounding context can significantly influence emotional responses, altering how individuals react to the agent's actions. Finally, the observations available to the agent may correspond to multiple possible models of the environment. Therefore, the agent must infer which environment model best explains observed behavior to act effectively.

To address these challenges, this project employs a Reinforcement Learning approach in a fully observable setting, assuming that the agent can perfectly detect human emotional states. In parallel, an ASP-based reasoning framework analyzes collected observations over time and infers a set of plausible environment models, producing a ranked list according to their likelihood. This integration allows the agent to combine experiential learning with symbolic reasoning, improving its ability to operate in complex and uncertain human-centered environments.

Moreover, the proposed model considers the possibility of dynamic changes in the environment over time, allowing the agent to adapt and compare its performance against a generic context-unaware baseline.

2 General Description of Our Approach (Interactions)

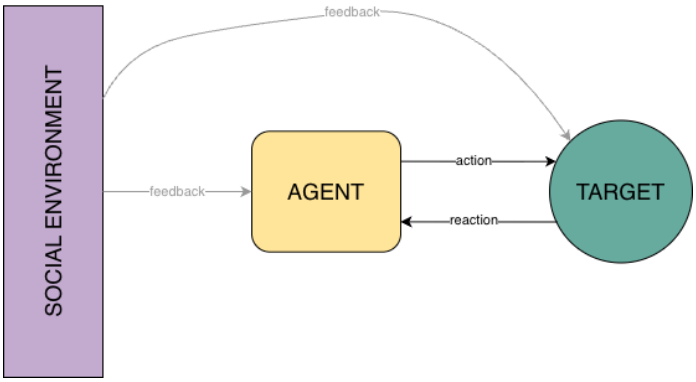


Figure 1: General schema of interaction between agent, human target, and environment.

The agent, human target, and environment form a closed-loop system that unfolds over discrete time steps as depicted in Figure 1. At each step:

- The **agent** observes the current emotional state of the human target and selects an action aimed at influencing it. Actions are chosen according to a learned policy derived from reinforcement learning.
- The **environment** provides feedback based on its current type and the agent's action. This feedback influences both the agent's learning and the human target's emotional state.
- The **human target** updates their internal emotional state in response to the agent's action and the environmental context, producing the next observation for the agent.

3 Human Model and Environment (Vector Formulation)

Modeling human emotional states is inherently challenging due to both inter-individual variability and contextual influences. In this work, we consider four core emotions (happiness, sadness, anger, and anxiety) which are used to represent the internal state of a human agent.

Our goal is to model a population of individuals that react differently to the same stimuli while preserving a consistent underlying behavioral structure. In particular, actions are assumed to have an intrinsic qualitative effect (positive, negative, or neutral), which is shared across individuals. However, each individual assigns a different magnitude (or weight) to these effects, resulting in heterogeneous but coherent behaviors.

This design choice allows us to capture two key aspects of human interaction: (i) general agreement on the nature of actions (e.g., a joke is typically perceived as positive, while insults are negative), and (ii) variability in emotional sensitivity across individuals.

Formally, the emotional state of the i-th individual at time t is represented as:

$$\mathbf{s}_t^{(i)} = \begin{bmatrix} s_{t,\,happiness}^{(i)} & s_{t,\,sadness}^{(i)} & s_{t,\,anger}^{(i)} & s_{t,\,anxiety}^{(i)} \end{bmatrix}^\top.$$

Each action a_t induces an emotion-dependent variation vector for individual i:

$$\boldsymbol{\delta}_{emotion}^{(i)}(a_t) = \begin{bmatrix} \delta_{happiness}^{(i)}(a_t) & \delta_{sadness}^{(i)}(a_t) & \delta_{anger}^{(i)}(a_t) & \delta_{anxiety}^{(i)}(a_t) \end{bmatrix}^\top,$$

where the sign of each component reflects the intrinsic nature of the action (positive, negative, or neutral), while its magnitude is sampled to model individual differences.

Each individual is initialized with a personalized set of action-dependent emotional variations. For each action $a \in \mathcal{A}$, the components are sampled as follows:

$$\delta_{emotion}^{(i)}(a) \sim \begin{cases} \mathcal{D}^+ & \text{if the action has a positive effect on } e \\ \mathcal{D}^- & \text{if the action has a negative effect on } e \\ \mathcal{D}^0 & \text{if the action has a neutral effect on } e \end{cases}$$

where $\mathcal{D}^+$, $\mathcal{D}^-$, and $\mathcal{D}^0$ are distributions over positive, negative, and near-zero values, respectively.

This initialization strategy ensures that all individuals share a consistent interpretation of actions (in terms of sign), while exhibiting different sensitivities due to the stochastic sampling of magnitudes.

The update rule for the emotional state of individual i is defined as:

$$\mathbf{s}_{t+1}^{(i)} = f(\mathbf{s}_t^{(i)}, a_t) = \boldsymbol{\delta}_{emotion}^{(i)}(a_t) \odot \exp\Big(-\frac{\mathbf{s}_t^{(i)}}{2}\Big) + \boldsymbol{\delta}_{environment}(t),$$

where $\odot$ denotes the element-wise (Hadamard) product and the exponential function is applied element-wise to the vector $\mathbf{s}_t^{(i)}$.

Two important aspects are captured by this formulation:

1. **State-dependence of actions:** The effect of an action depends on the current emotional state. For instance, the same joke might increase happiness more when the individual is neutral than when they are already very happy. (This will support our use of RL formulation with Q-learning)

2. **Attenuated effect on high-level emotions:** Emotions that are already at high levels are more resistant to change. The exponential term ensures that actions have a diminishing effect as the corresponding emotion increases, modeling the difficulty of "shifting" strong emotions.

This formulation aligns with the implementation in code, where each emotion is updated independently but scaled by its current value and modulated by both the action and the environment.

An example of the resulting state transition behavior is illustrated in Figure 2, where the influence of the environment and the dependence on the current emotional state are highlighted.

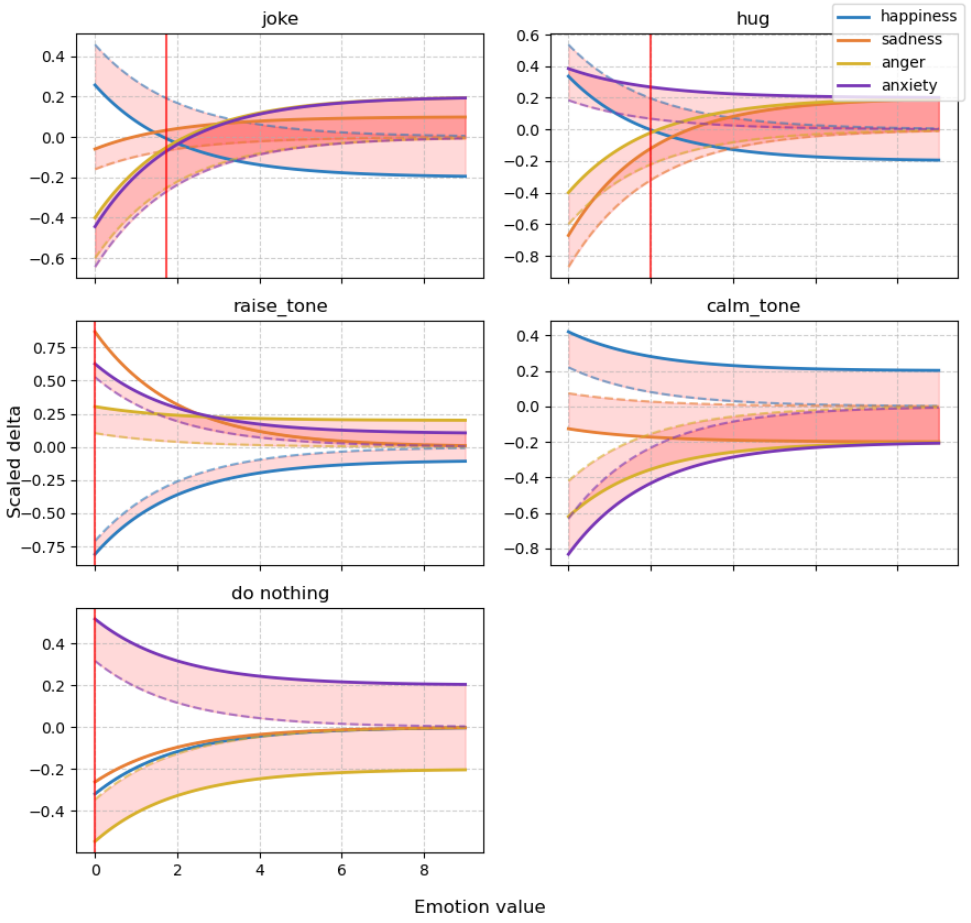


Figure 2: State transition function of a sample human. The red gap denotes the environment-induced shift, while the red line highlights the threshold beyond which the action has a negative effect on the happiness.

4 Agent Modeling: Reinforcement Learning Approach

The decision-making problem is modeled using a standard Reinforcement Learning (RL) framework, specifically Q-learning. At each time step t, the agent observes the emotional state of an individual, represented by the vector $\mathbf{s}_t^{(i)}$, and selects an action $a_t \in \mathcal{A}$.

The goal of the agent is to learn a policy $\pi : \mathbf{s} \to a$ that maximizes long-term positive emotional outcomes. In particular, the agent aims to increase happiness while minimizing negative emotions such as sadness, anger, and anxiety.

Since the emotional state $\mathbf{s}_t^{(i)} \in R^4$ is continuous, the state space is discretized internally. This allows the use of tabular Q-learning, which is well-suited for environments with relatively low-dimensional representations and enables effective exploration of the state-action space.

Reward Function. The reward function is designed to capture both the direct emotional impact on the human and the feedback induced by the environment. Given the variation in the human emotional state $\boldsymbol{\delta}_{human}^{(i)}$ and the environment feedback $\boldsymbol{\delta}_{environment}$, the total reward is defined as:

$$R = R_{human} + R_{environment}.$$

The human-related component is computed as a weighted sum of emotional variations:

$$R_{human} = \sum_{e \in \mathcal{E}} w_e^{human} \cdot \delta_e^{human,(i)},$$

where $\mathcal{E} = \{\text{happiness}, \text{sadness}, \text{anger}, \text{anxiety}\}$ and the weights reflect the relative importance of each emotion. In particular, positive emotions such as happiness are strongly rewarded, while negative emotions are penalized.

Similarly, the environment contribution is defined as:

$$R_{environment} = \sum_{e \in \mathcal{E}} w_e^{env} \cdot \delta_e^{env}.$$

In the default configuration, the weights are set as:

$$w_{happiness}^{human} = 100, \quad w_{sadness}^{human} = w_{anger}^{human} = w_{anxiety}^{human} = -10,$$

$$w_{happiness}^{env} = 50, \quad w_{sadness}^{env} = w_{anger}^{env} = w_{anxiety}^{env} = -5.$$

This weighting empirically has given good results. This formulation allows the agent to balance immediate emotional impact with contextual feedback, encouraging actions that are both individually beneficial and contextually appropriate.

Due to intrinsic randomness in both initial emotional states and individual-specific reaction parameters $\boldsymbol{\delta}_{emotion}^{(i)}(a)$, the environment exhibits stochastic behavior in different episodes. To account for this variability, the training procedure was modified as follows: multiple episodes were simulated in various sampled individuals, and the resulting Q-tables were averaged.

This approach allows the agent to approximate the expected effect of actions over a population of heterogeneous individuals, capturing typical behavioral patterns while reducing the influence of outliers.

The coverage of the discretized state space achieved during training is illustrated in Figure 3, showing that the agent explores a large portion of the state space across episodes. Effective learning is further evidenced by the convergence of the Q-table, as

shown in Figure 4, where the mean variation of the Q-values decreases and stabilizes over the episodes.

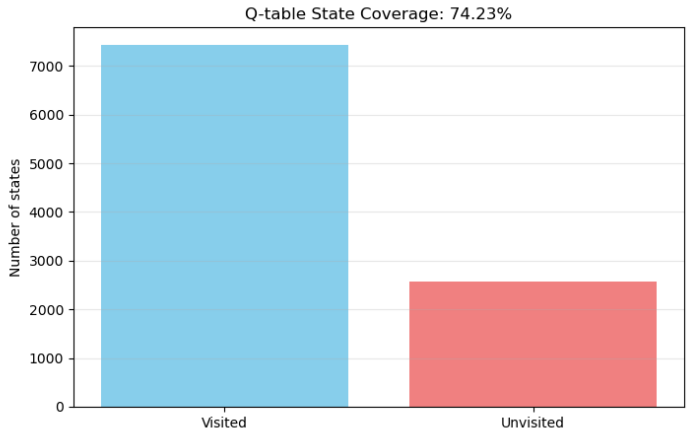


Figure 3: Coverage of the discretized state space by the Q-table. Training achieves good exploration across all episodes.

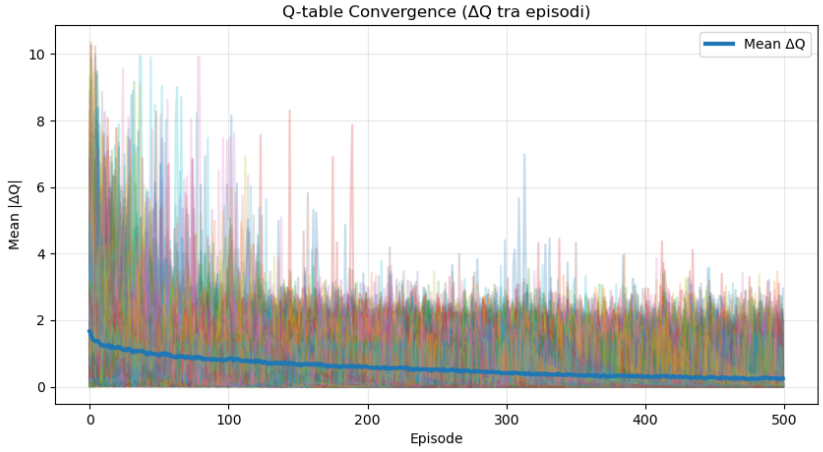


Figure 4: Q-learning convergence approaching the optimal solution, demonstrating effective learning.

5 ASP Problem Modeling and Environment Inference

While Reinforcement Learning is used to learn optimal behaviors within a given environment, it does not explicitly address the problem of identifying which environment the agent is currently interacting with. In human-centered scenarios, however, contextual information plays a crucial role: the same action may lead to very different outcomes depending on the surrounding environment.

For this reason, we employ Answer Set Programming (ASP) as a symbolic reasoning layer on top of the RL component. ASP is particularly well-suited for this

task for several reasons. First, it allows us to explicitly encode domain knowledge in terms of logical rules and constraints, enabling interpretable reasoning over observations. Second, it naturally supports reasoning under incomplete information, allowing the system to infer plausible environments even when some observations are missing. Third, it provides a flexible mechanism for handling context switching, enabling the agent to dynamically adapt its behavior by selecting the most appropriate environment model. Finally, by separating environment inference from policy learning, ASP allows each RL policy to be trained for a specific context, improving overall robustness and modularity.

Answer Set Programming is therefore used as the inference engine for identifying the most likely environment based on observed human behaviors. Each environment is defined by a set of required features and associated scoring rules, reflecting how strongly specific observations support or contradict a particular environment. Invalid environments are eliminated when required observations are missing.

The inference process consists of the following steps:

1. **Observation Matching:** The agent collects observations about human behaviors, such as clothing, language, voice, personal space, facial expressions, posture, and ambient noise.

2. **Validation:** Any environment whose required observations are not met is marked as invalid:

```
invalid(E) :- required(E,X), not observed(X).
```

3. **Scoring:** Each valid environment receives a score for each observed feature. Positive or negative values reflect how consistent the observation is with the environment:

```
score(Environment, Feature, Value).
```

4. **Total Score Computation:** The total score for an environment is computed as the sum of all individual feature scores:

```
total_score(E,Total) :- Total = #sum {S,X : observed(X),
                        score(E,X,S)}.
```

5. **Candidate Selection:** Environments with a positive total score are considered possible candidates:

```
possible(E) :- total_score(E,T), T > 0.
```

6. **Contribution Analysis:** Each observation's contribution can be categorized as positive or negative:

```
positive(E,X,S) :- contribution(E,X,S), S > 0.
negative(E,X,S) :- contribution(E,X,S), S < 0.
```

7. **Best Environment Selection:** The environment with the highest total score is selected as the best explanation for the current observations:

```
best(E) :- total_score(E,T), T = #max{T1,E1 :
           total_score(E1,T1), T1 > 0}.
```

This framework allows the agent to reason symbolically over observations and dynamically infer the most plausible environment in real time, providing both a *best guess* and a set of possible alternatives ranked by plausibility. This tight integration between learning and reasoning is a key aspect of the proposed approach.

6 Results and Baseline Comparison

To evaluate the effectiveness of our ASP-powered context-switching RL agent, we compare it against a baseline RL agent trained purely with Q-learning, without any context inference. The baseline agent interacts with the same human-environment setup but is aware only of a fixed context and cannot adapt to changes based on observations.

Our experimental hypotheses are as follows:

- The RL agent, even without context awareness, is capable of generally increasing human happiness over time.
- The ASP-enabled adaptive agent can better infer environmental changes and adjust its actions accordingly, leading to faster and more effective increases in happiness.
- The adaptive agent can also reduce negative emotions more efficiently than the context-unaware baseline.

Experimental results confirm these expectations as shown in Figure 5. The adaptive model consistently outperforms the context-unaware baseline, except in cases where different contexts produce very similar human responses. As the environments diverge, the advantages of context-aware modeling become evident, highlighting the importance of combining learning and symbolic reasoning in human-centered settings. An example of this comparison is shown in Figure 6.

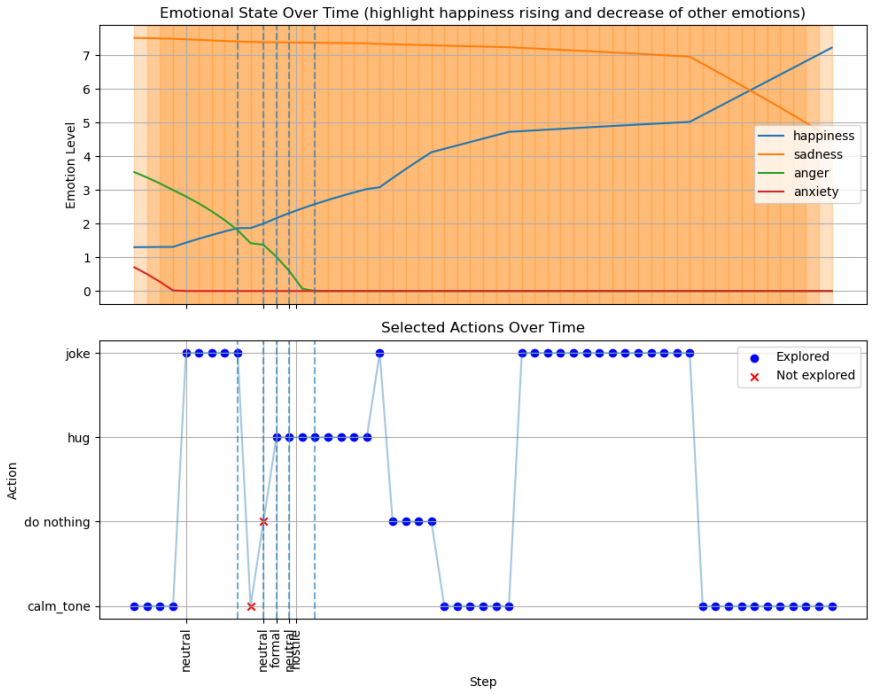


Figure 5: Example experiment showing the agent's performance. Orange areas indicate regions where happiness is maximized and negative emotions are minimized.

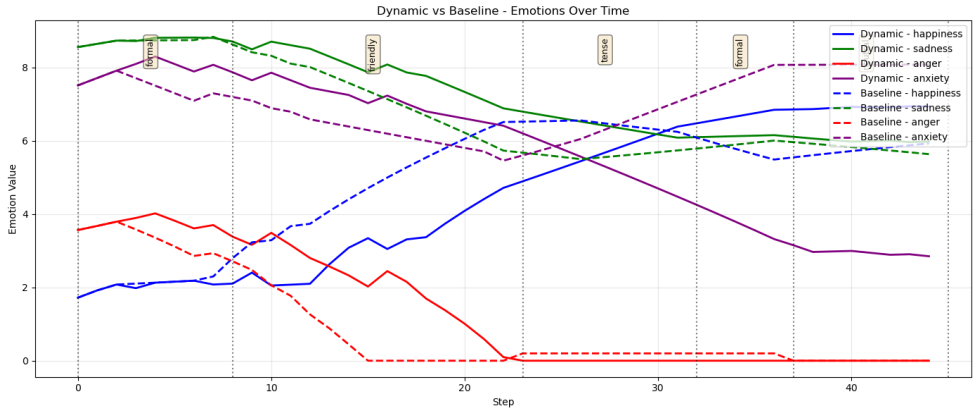


Figure 6: Comparison of context-aware versus context-unaware agent performance across different environments. The adaptive agent achieves higher happiness while reducing negative emotions more effectively.

7 Conclusion and Future possible extensions

Our experiments show that the adaptive, context-aware agent consistently outperforms the baseline context-unaware agent, particularly in dynamic and varying social contexts. By combining Reinforcement Learning with ASP, the agent can infer the environment and adjust its actions, resulting in higher human happiness and better management of negative emotions.

For future work, several enhancements could be implemented to further improve the agent:

- **Feature-based state representation:** Moving from discrete Q-tables to feature vectors would allow the agent to handle larger and continuous state spaces, enabling more fine-grained modeling of human emotions and interactions.
- **Advanced ASP optimization:** Using `#maximize` statements or other ASP optimization techniques would allow more direct and flexible reasoning over multiple candidate environments, improving inference efficiency and expressiveness.
- **Contextual combination of observations:** Incorporating combinations of observations (e.g., facial expression plus posture) or weighted indicators could capture more nuanced environmental cues, allowing the agent to detect subtle contextual shifts and adapt more accurately.
- **Partial observability:** Extending the framework to partially observable human states would increase realism, simulating scenarios where the agent cannot perfectly perceive the human's internal state, and requiring probabilistic inference or belief-state tracking.
- **Dynamic environment adaptation:** Implementing mechanisms to handle sudden or continuous changes in the environment over time would make the agent more robust and capable of lifelong adaptation, beyond fixed time windows.
- **Detection of wrong context inference:** The agent could monitor its performance in the currently inferred environment and, if results are poor, switch

to an alternative environment from the ranked list of candidates. This would simulate adaptation and dynamic behavior over time.

- **Explainability:** ASP can be leveraged to provide a clear explanation for why a specific environment was inferred and why the corresponding policy was chosen, improving interpretability of the agent's decisions.
- **A more realistic scenario:** In a future extension, the agent could infer human emotions from natural language interactions, allowing for richer and more realistic human-agent interactions. While challenging, this approach would better capture real-life social dynamics.

Implementing these improvements would enhance the agent's adaptability, reasoning capabilities, and realism, further demonstrating the potential of combining symbolic reasoning with learning-based approaches in complex human-centered environments.

Together, these improvements would enhance the agent's adaptability, scalability, and reasoning capabilities, demonstrating the full potential of integrating symbolic reasoning with learning-based approaches in complex, human-centered environments.